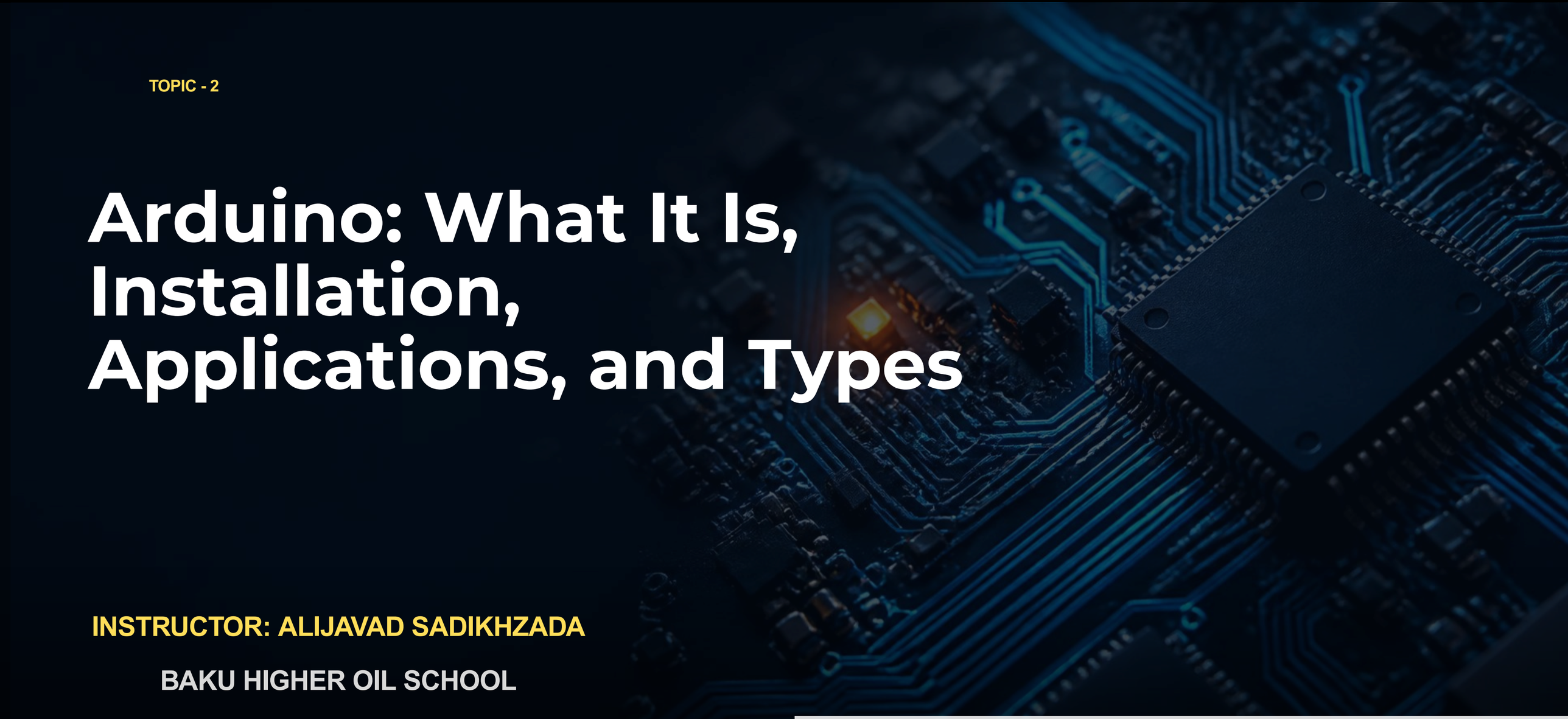
TOPIC - 2

Arduino: What It Is, Installation, Applications, and Types

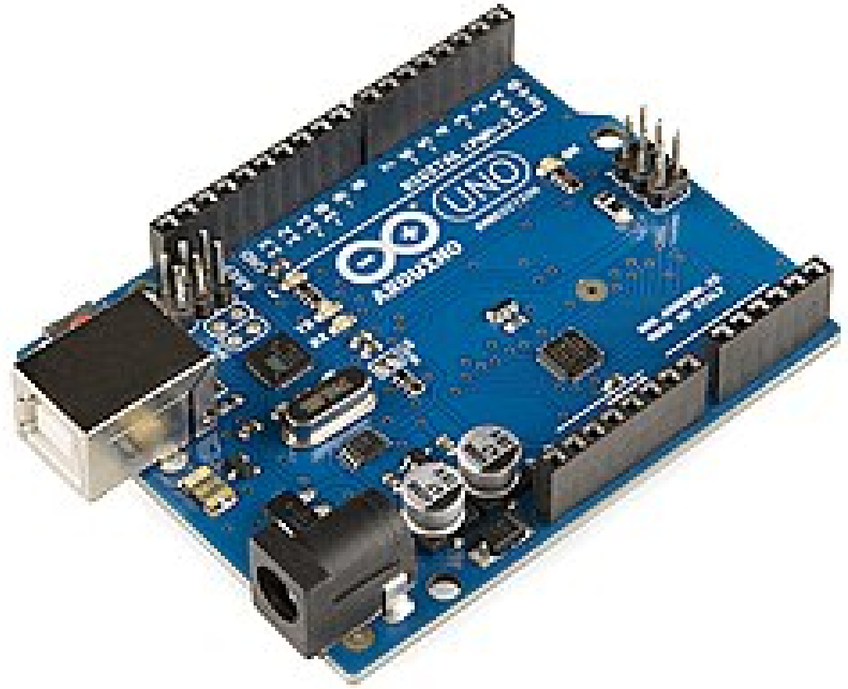
INSTRUCTOR: ALIJAVAD SADIKHZADA

BAKU HIGHER OIL SCHOOL

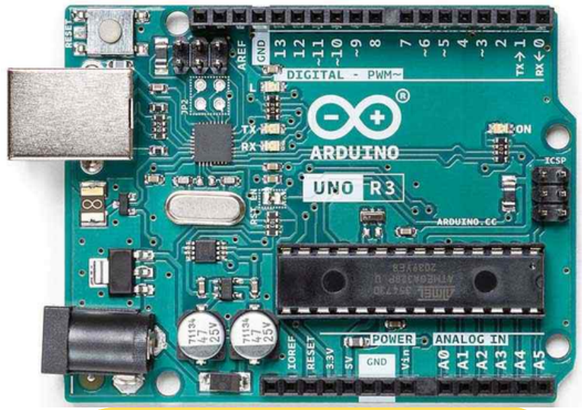


BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

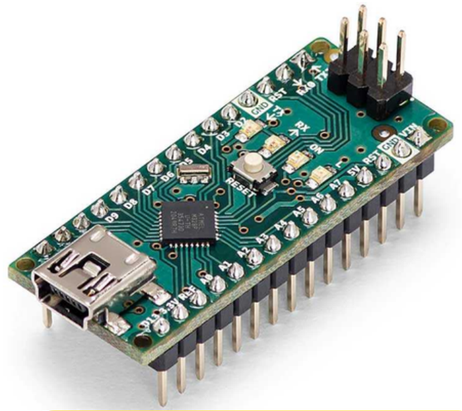
Arduino is an open-source electronics platform based on easy-to-use hardware and software. It is designed for anyone who wants to create interactive projects. Arduino can receive data from various sensors to detect its environment and interact with it by controlling lights, motors, and other actuators.



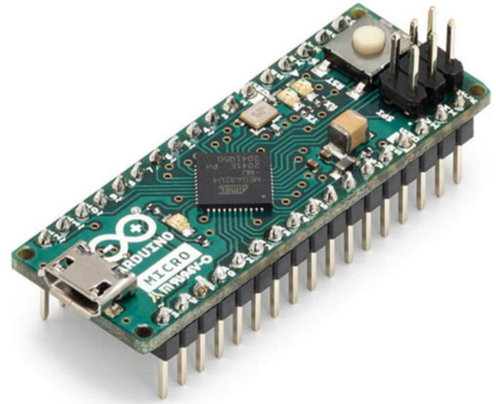
WHAT IS ARDUINO?



ARDUINO UNO R3



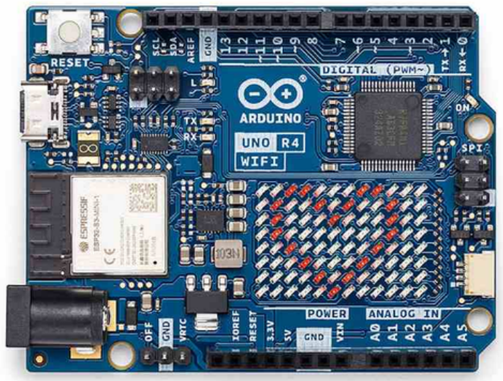
ARDUINO NANO



ARDUINO MICRO



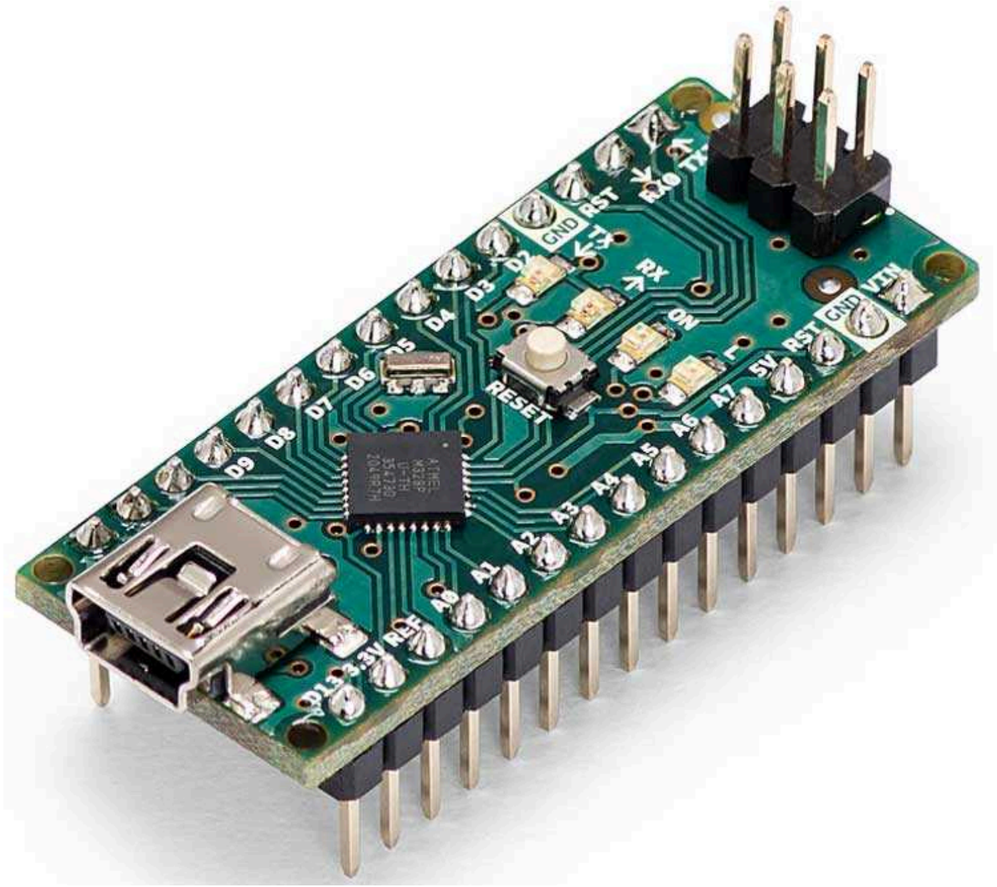
ARDUINO MEGA2560
REV3



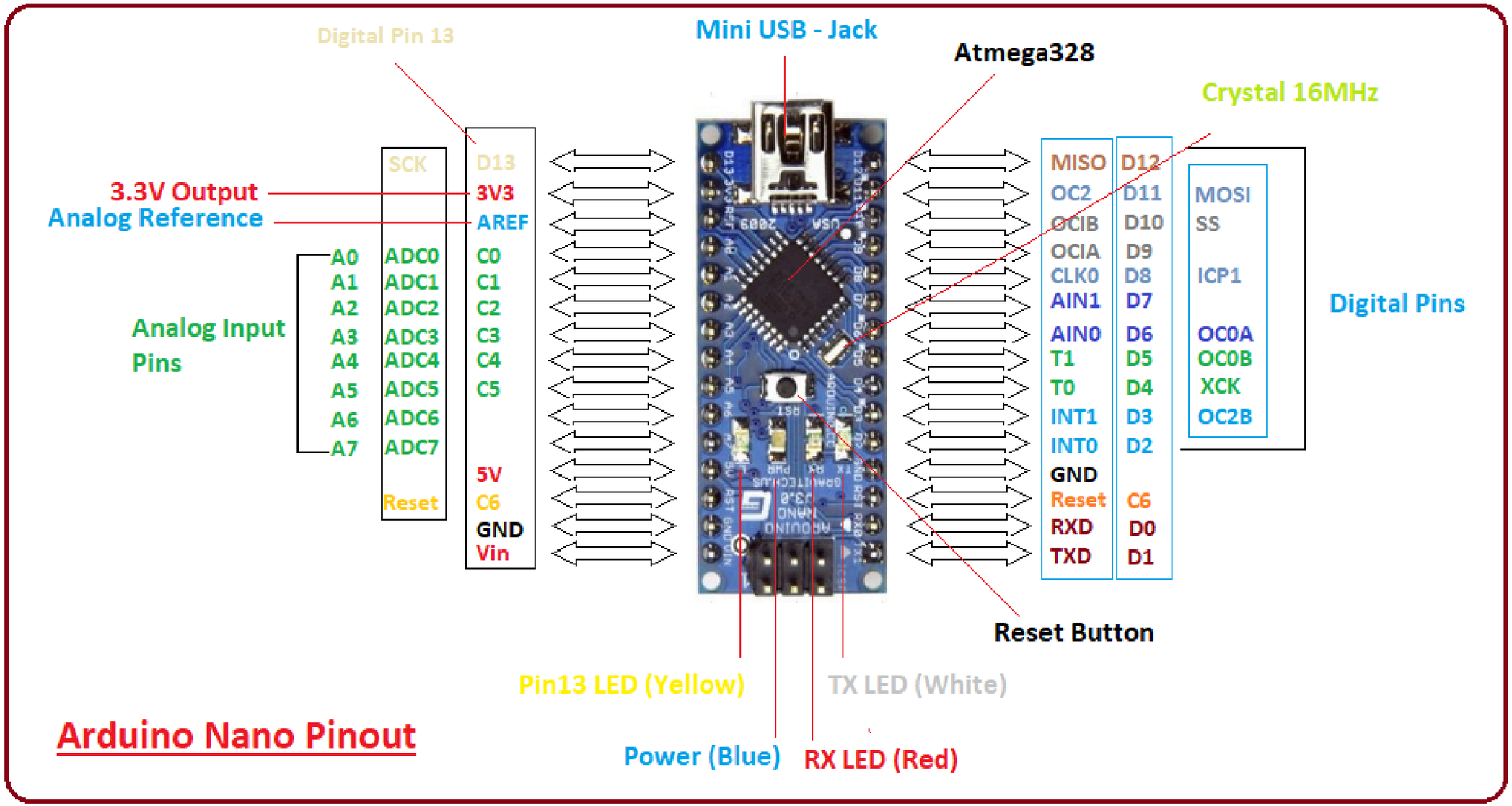
ARDUINO UNO R4 WIFI

COMMON ARDUINO BOARDS

- 16 MHz Clock Speed
- 32 KB Flash Memory
- 2 KB SRAM
- 14 Digital I/O Pins
- 8 Analog Input Pins, with 10-bit resolution
- ATmega328P Microcontroller
- 1 UART Communication Protocol
- 1 SPI Communication Protocol
- 1 I²C Communication Protocol

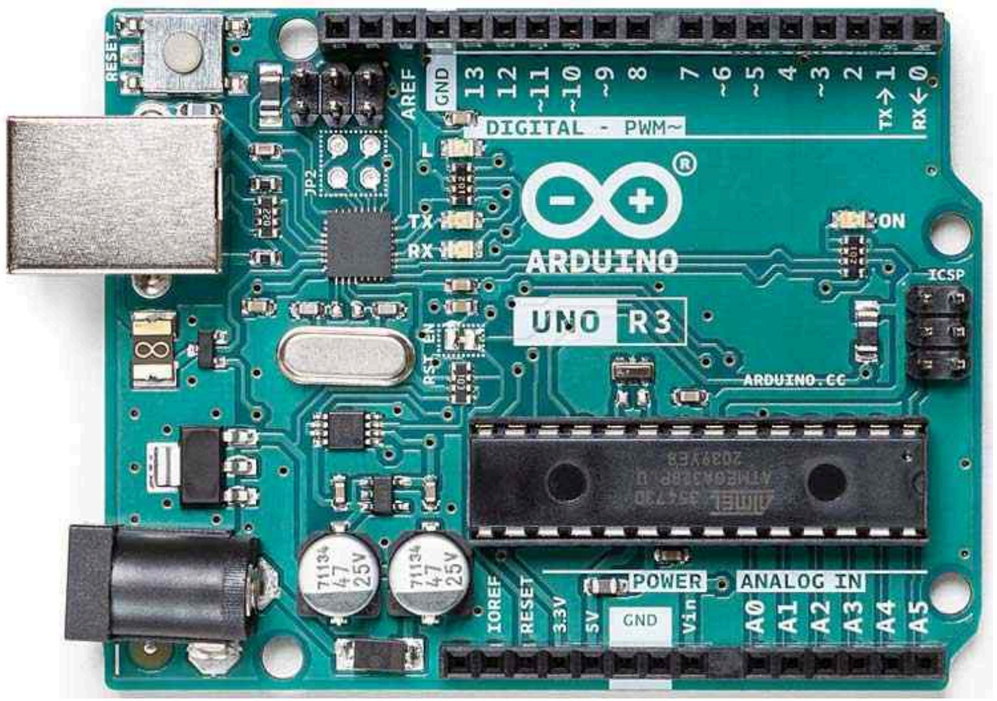


ARDUINO NANO

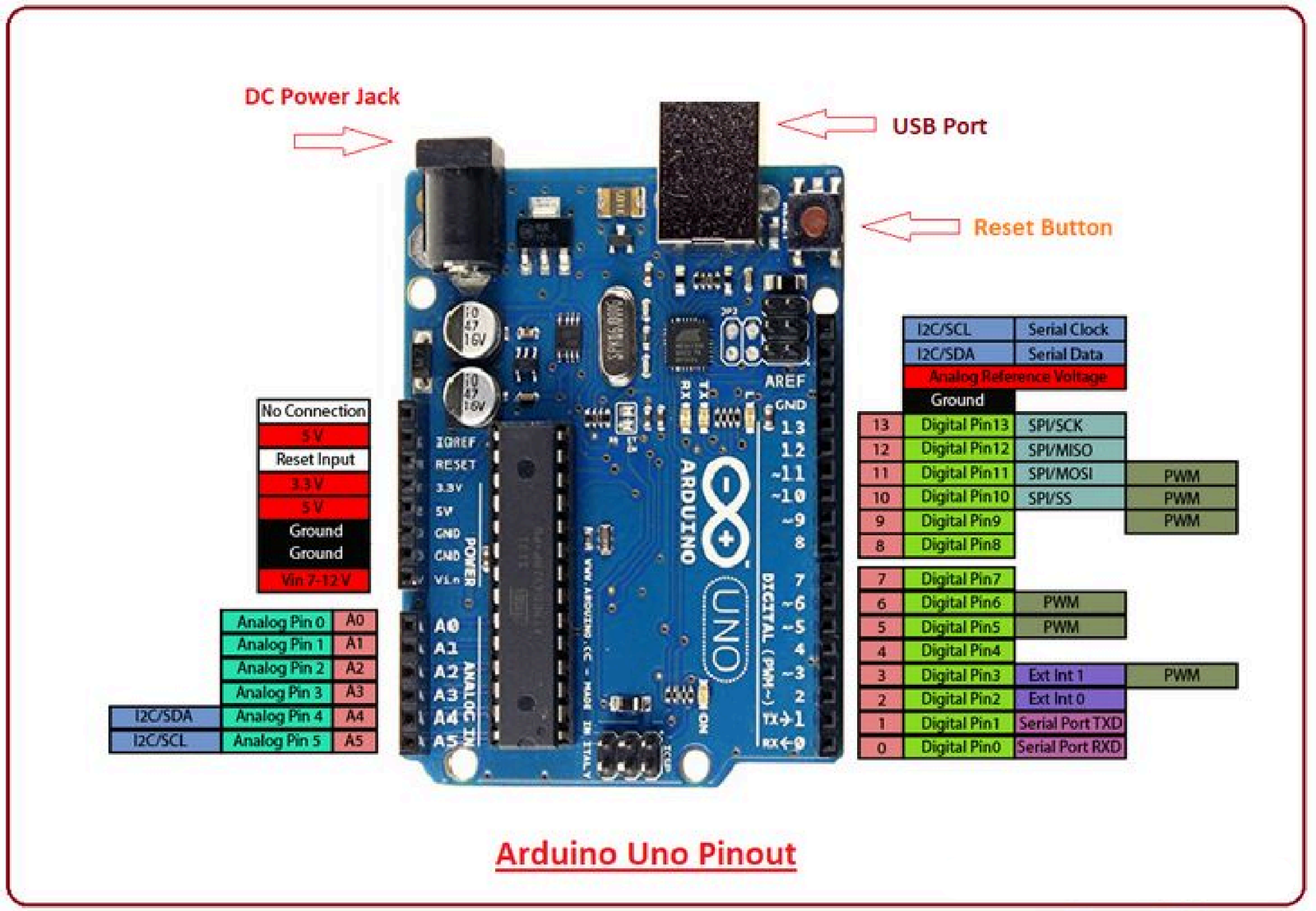


ARDUINO NANO PINOUT

- 16 MHz Clock Speed
- 32 KB Flash Memory
- 2 KB SRAM
- 14 Digital I/O Pins
- 6 Analog Input Pins, with 10-bit resolution
- ATmega328P Microcontroller
- 1 UART Communication Protocol
- 1 SPI Communication Protocol
- 2 I²C Communication Protocols



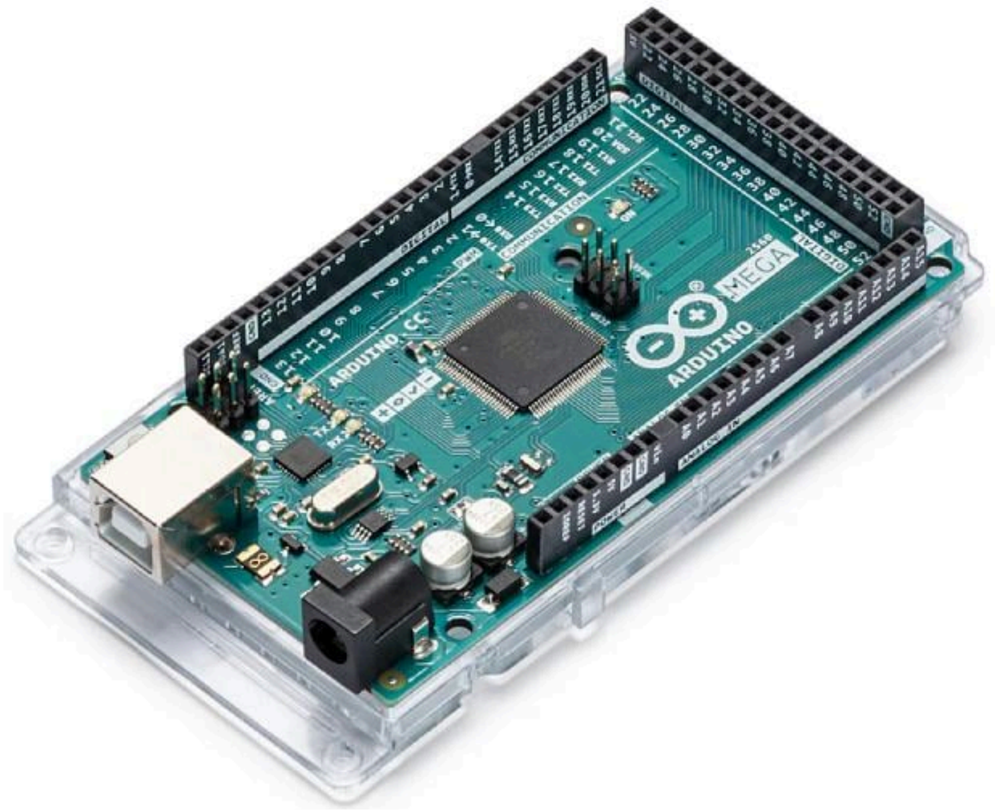
ARDUINO UNO



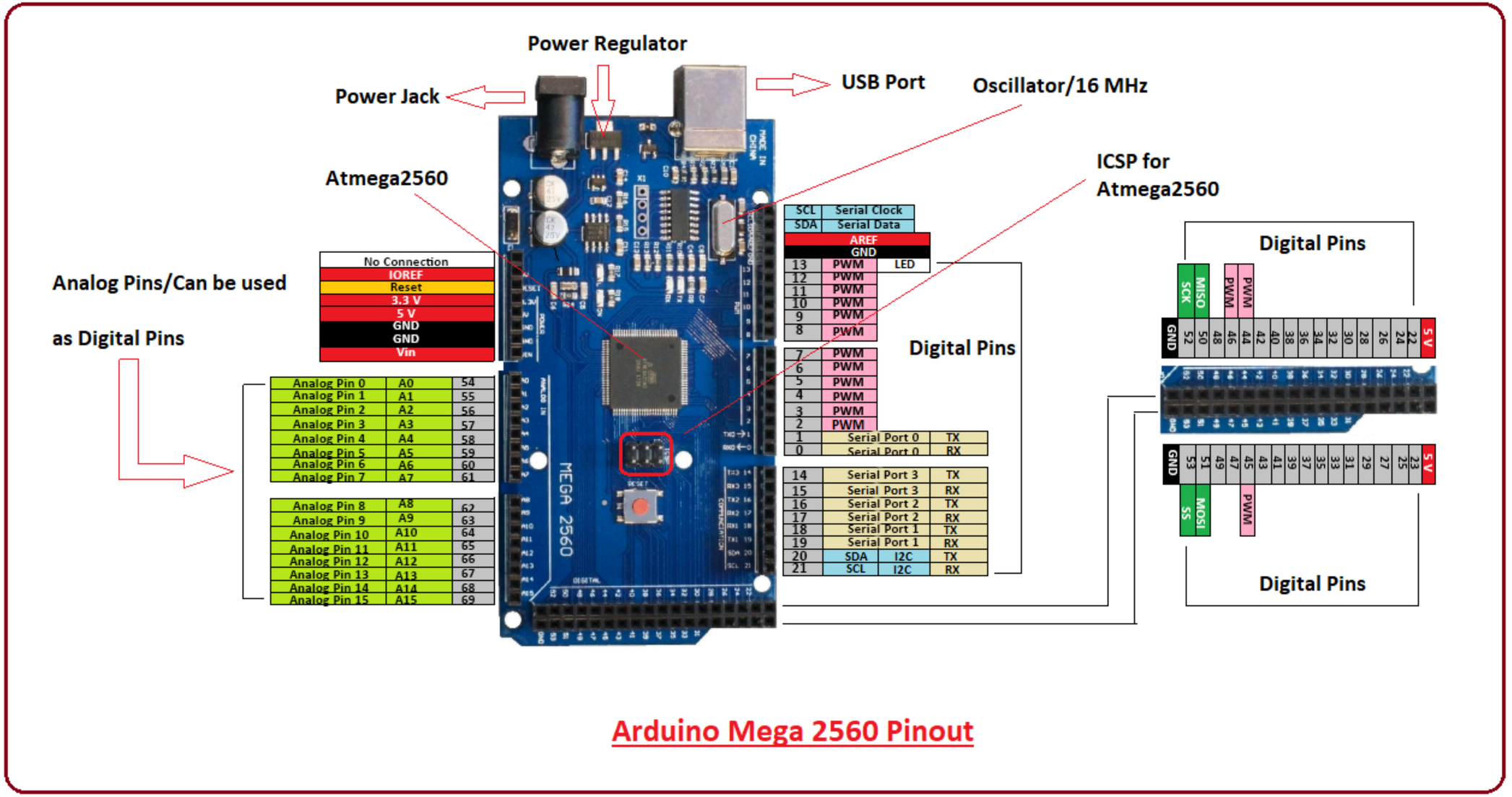
Arduino Uno Pinout

ARDUINO UNO PINOUT

- 16 MHz Clock Speed
- 256 KB Flash Memory
- 8 KB SRAM
- 54 Digital I/O Pins
- 16 Analog Input Pins, with 10-bit resolution
- ATmega2560 Microcontroller
- 15 PWM Outputs
- 4 UART Communication Interfaces
- 1 SPI Communication Interface
- 1 I²C Communication Interface



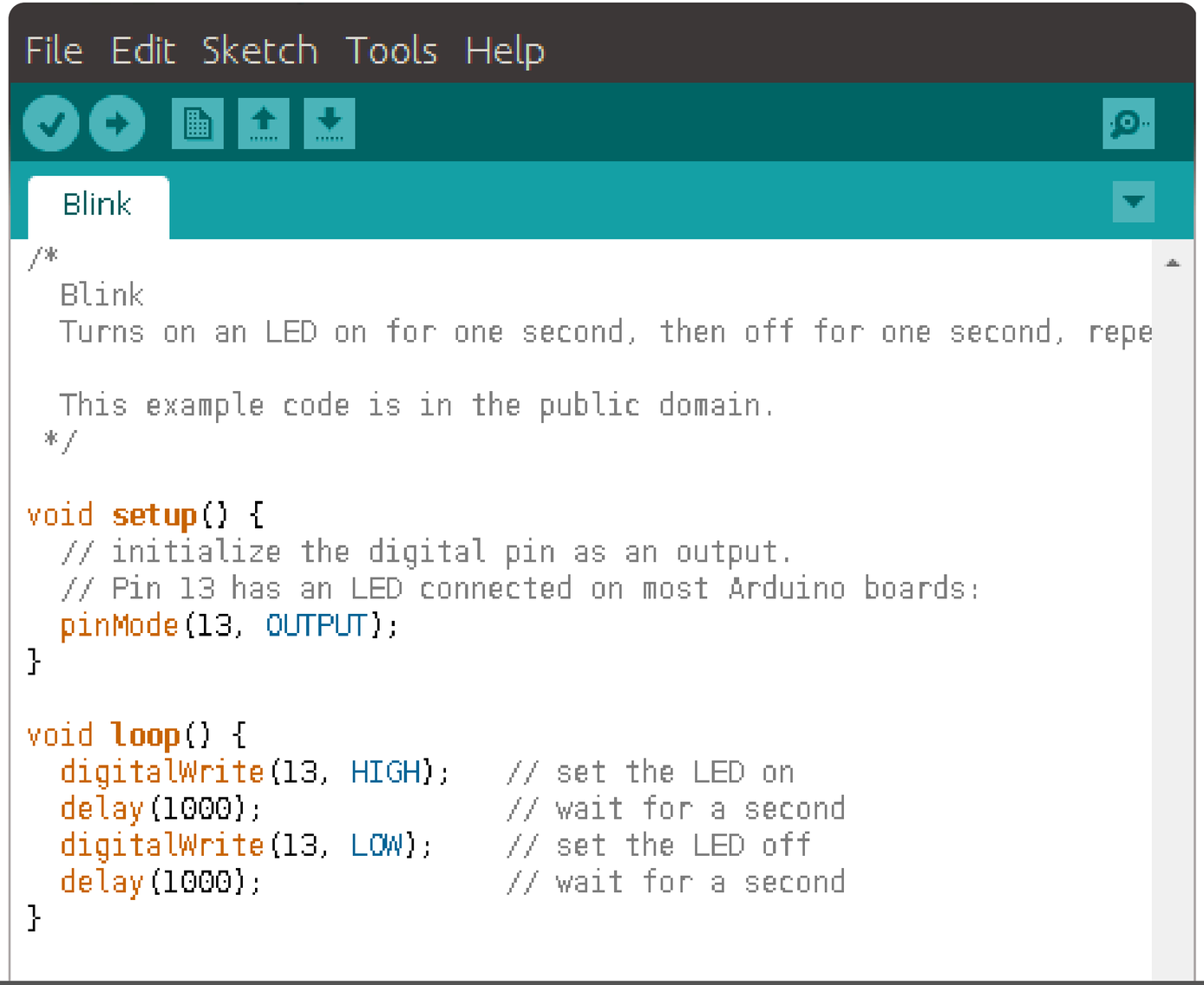
ARDUINO MEGA



Arduino Mega 2560 Pinout

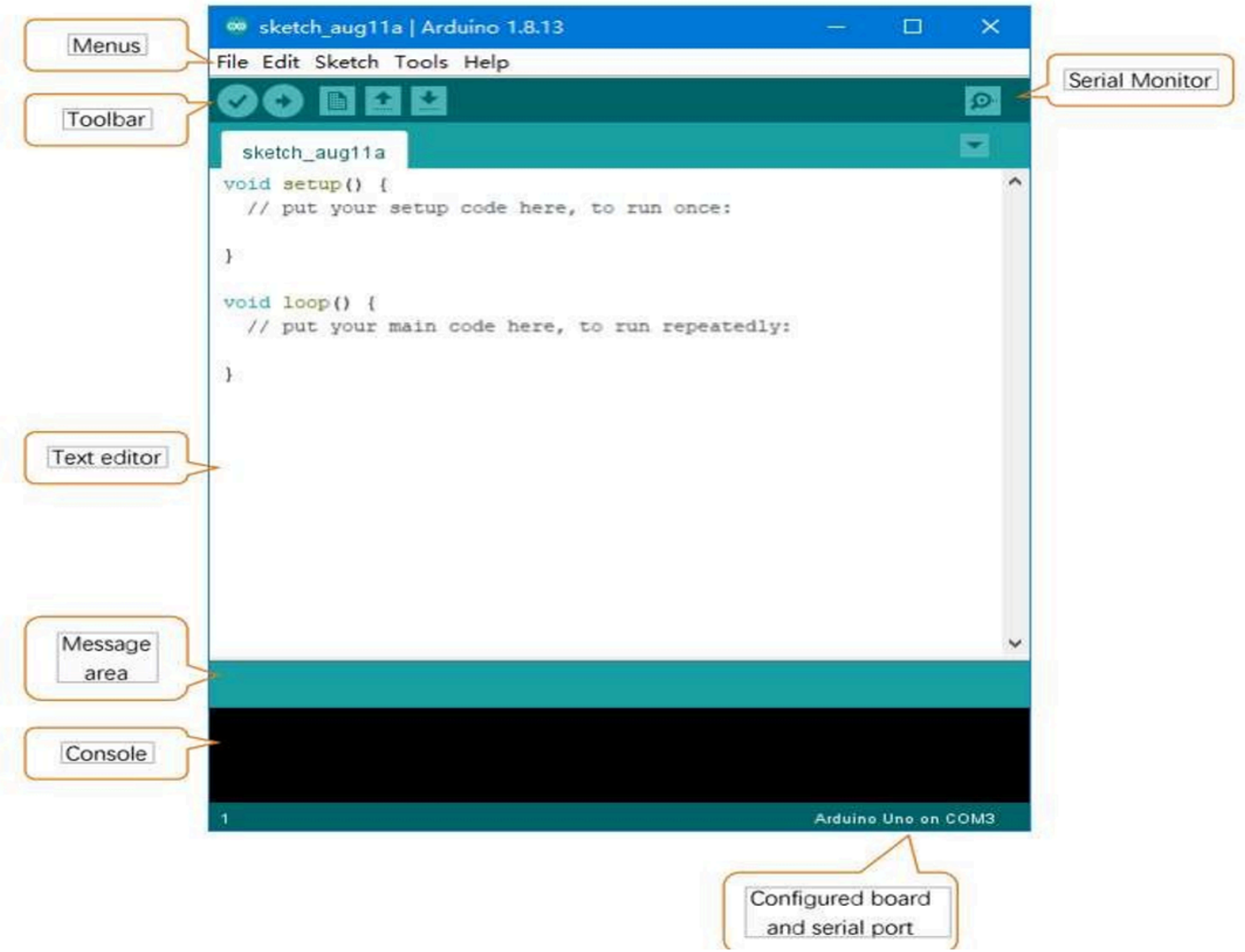
ARDUINO MEGA PINOUT

Arduino IDE is a cross-platform development environment originally written in Java. Its integrated development features allow users to program in C/C++ and upload programs to compatible boards.
It also provides debugging tools to identify and fix problems in the code.
Programmers can access various libraries, use syntax highlighting, communicate with the board through the Serial Monitor, and verify their code using the Arduino IDE.
As a bonus, the software can be used offline without an internet connection.



ARDUINO IDE

- **Main Menu**
- **Toolbar** – The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the Serial Monitor.
- **Text Editor** – Used to write your code.
- **Message Area** – Provides feedback during saving and exporting and displays errors.
- **Text Console** – Displays text output from the Arduino IDE, including detailed error messages and other information.
- **Configuration** – Shows the selected board and serial port in the bottom-right corner of the window.

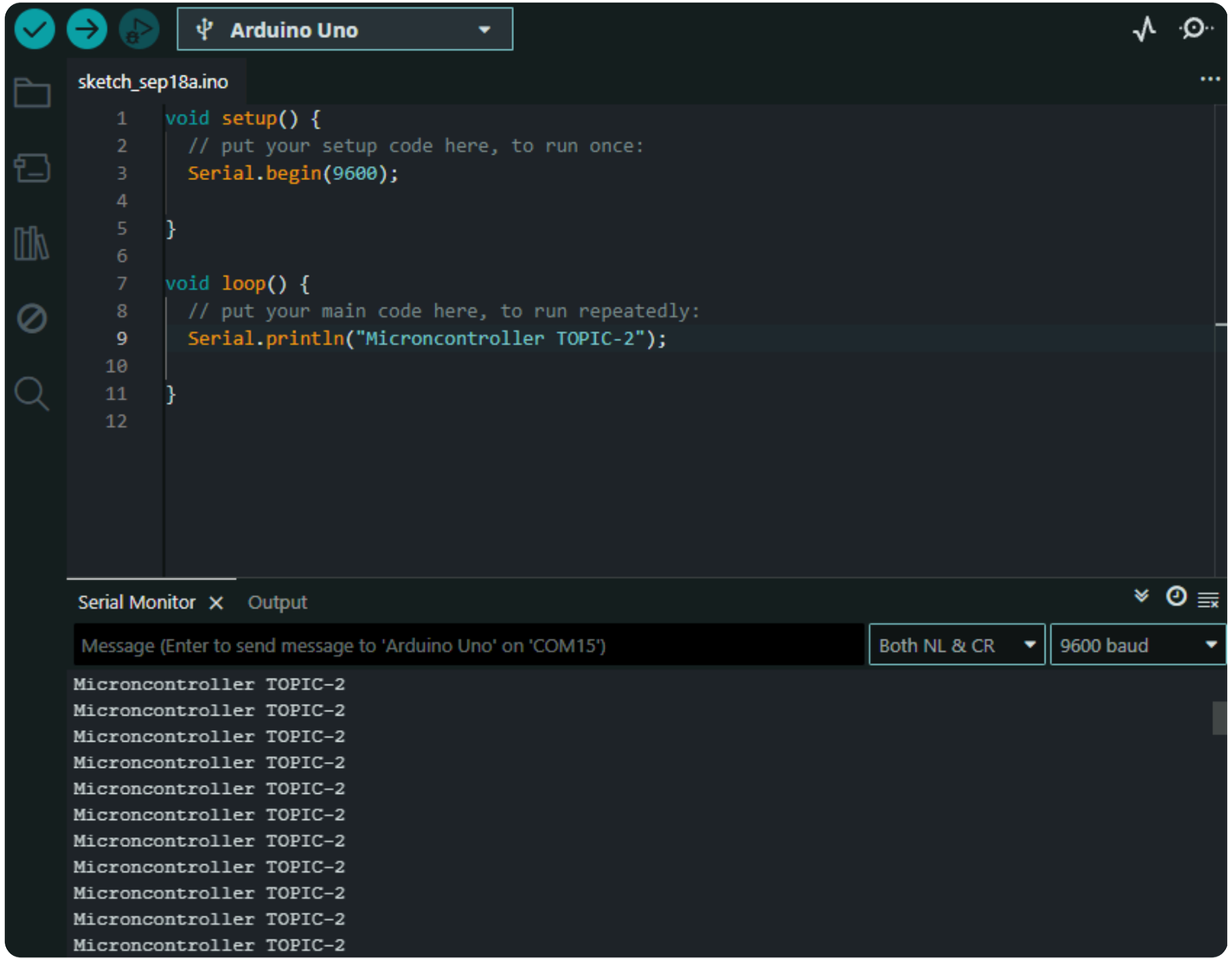


ARDUINO IDE

Serial.print();



- **Serial.begin(9600);** – Used to start serial communication. It is written inside the **void setup()** function. The value 9600 means that 9600 bits are sent and received per second.
- **Serial.print();** – Used to print something on the Serial Monitor. The element to be printed is written inside the parentheses. Text is written inside " ", while numbers and variables are written without quotation marks.
- **Serial.println();** – Performs the same function as Serial.print(), but moves to the next line after printing.



SERIAL PRINT

- Serial.print(“Microcontrollers Topic2”);
- Serial.print(5);

- Serial.print(“Microcontrollers Topic2”);

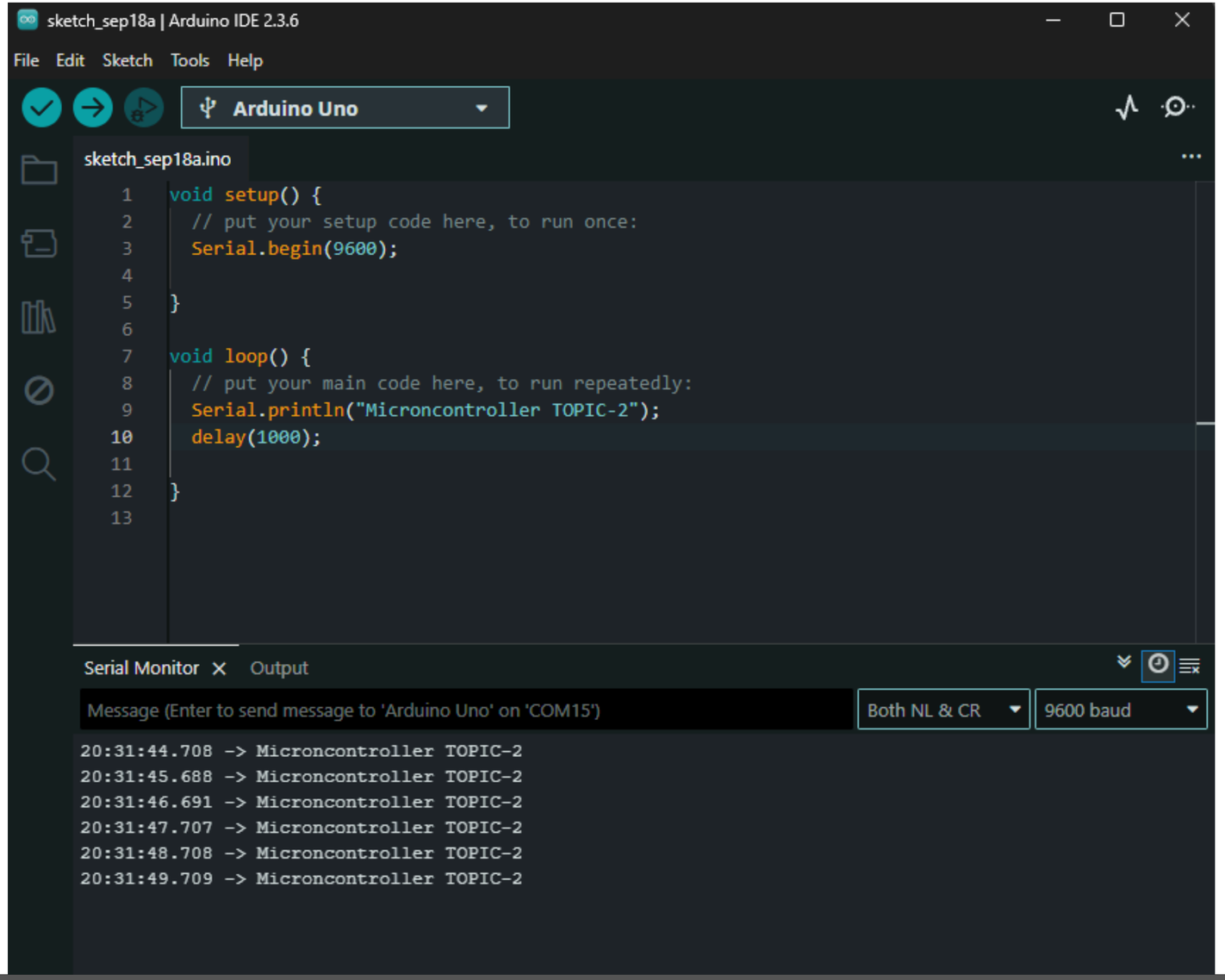
- Serial.print(“Microcontrollers Topic2”+String(5));

- Serial.print(“Microcontrollers”+”Topic2”);

- Serial.println(“Microcontrollers Topic2”);
- Serial.print(5);



- **delay();** – Pauses the program for a specified amount of time (in milliseconds). There are 1000 milliseconds in one second.
- During the delay, mathematical calculations and pin operations cannot be performed.
- **delayMicroseconds();** – Pauses the program for a specified amount of time (in microseconds). There are 1,000,000 microseconds in one second.



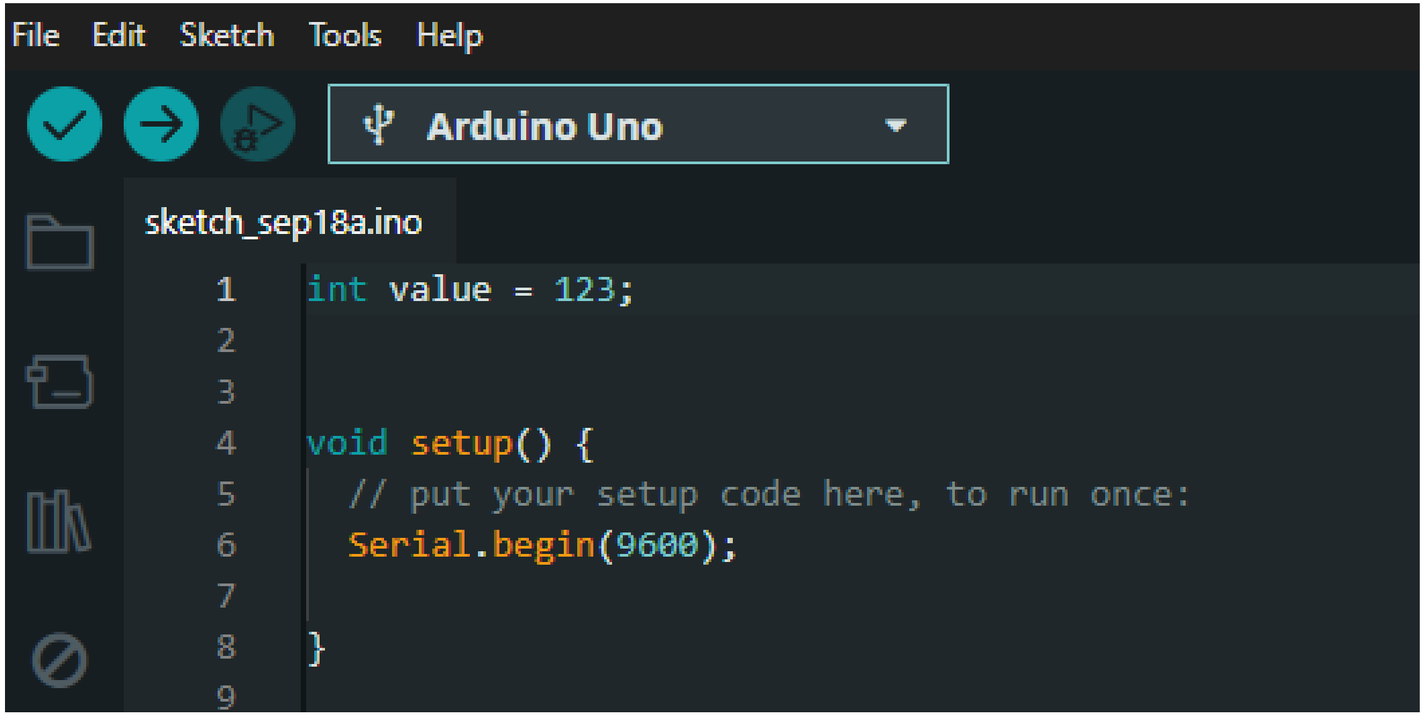
- **int** – Integer numbers without decimals.
- **char** – Character variables and letters.
- **String** – A sequence of characters or letters.
- **float** – Decimal numbers; can store up to 7 decimal digits of precision.
- **double** – Decimal numbers; can store up to 15 decimal digits of precision.
- **bool** – Stores true (1) or false (0) values.

Data type	Size(bytes)	Range
char	1	-128 to 127
unsigned char	1	0 to 255
short	2	-32,768 to 32,767
unsigned short	2	0 to 65535
int	4	-2147483648 to +2147483647
unsigned int	4	0 to 4294967295
long	4	-2147483648 to +2147483647
Unsinged long	4	0 to 4294967295
float	4	-3.4e-38 to +3.4e-38
double	8	1.7 e-308 to 1.7 e+308
long double	8	1.7 e-308 to 1.7 e+308
bool	1 bit	
void	-	-
wchar_t	2 or 4	1 wide character



DATA TYPES

- **int (Integer)** – Used to store whole numbers. Its size is usually 4 bytes, allowing it to store values from -2,147,483,648 to 2,147,483,647.



- **Char** – Used to declare character variables. A character variable can store only one character.



CHARACTER

- **String** – Used to store text or a sequence of characters in programs.

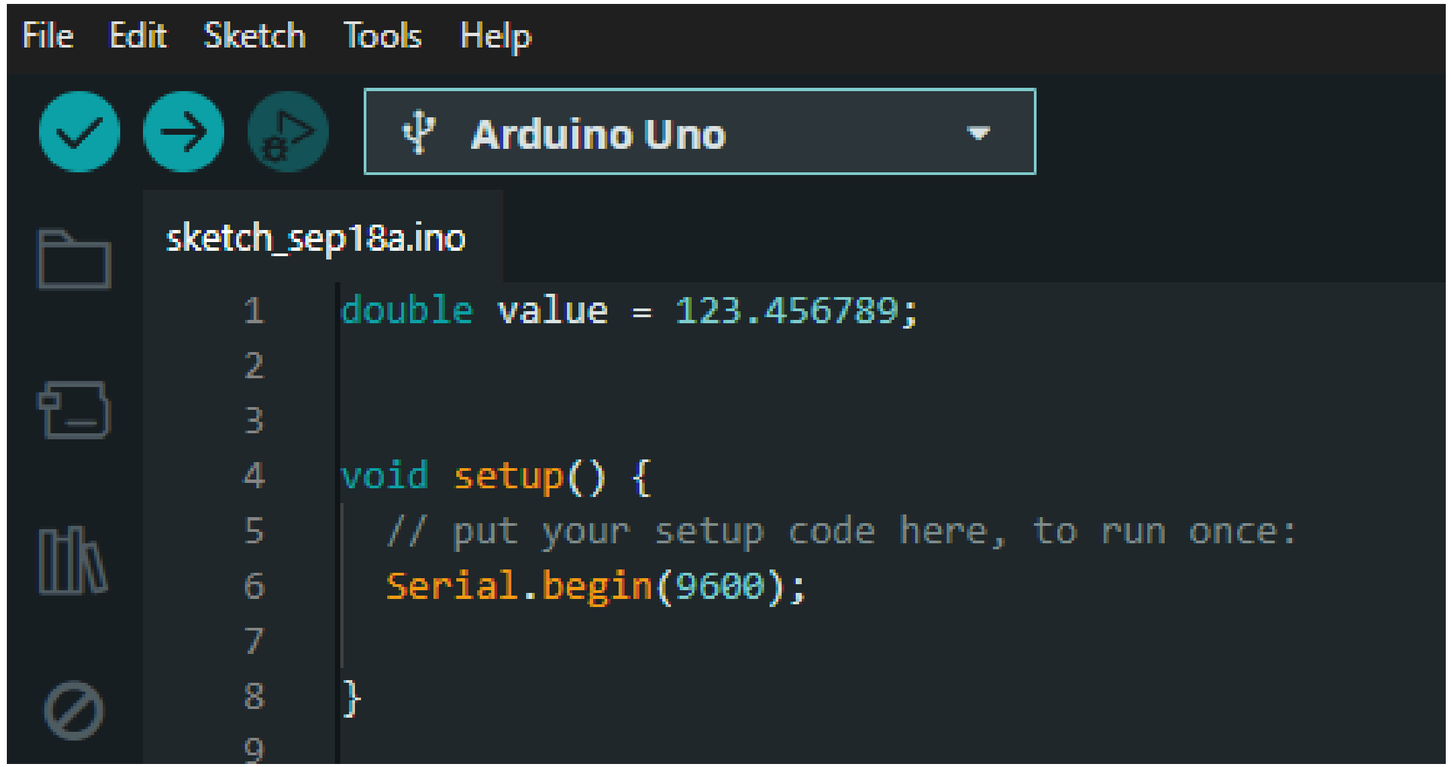


- **Float** – A variable type used to store decimal numbers with up to 7 decimal digits of precision.

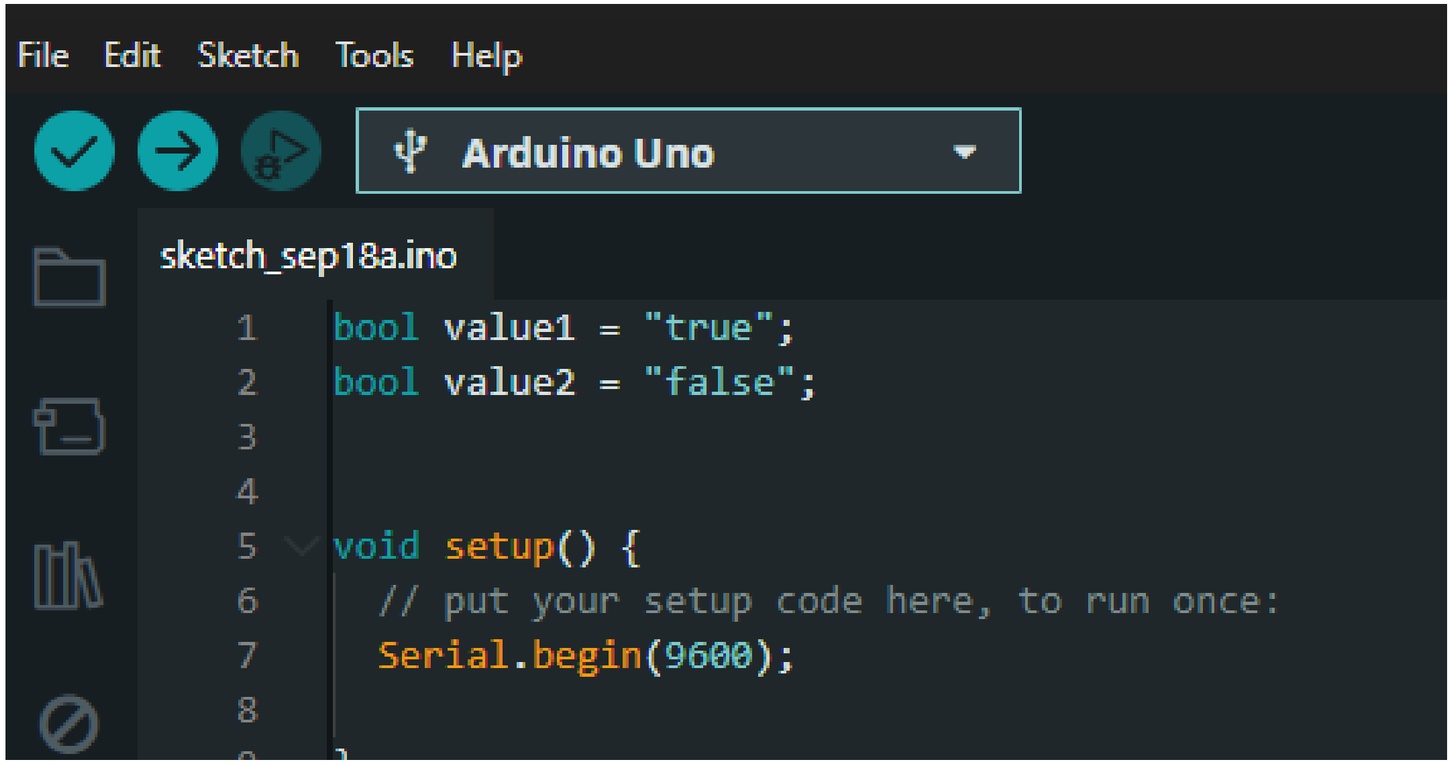


FLOAT

- **Double** – A variable type used to store decimal numbers with up to 15 decimal digits of precision.



- **Bool** – A variable type that can have either true (1) or false (0) values.



- **+** – Adds two values (value1 + value2)
- **−** – Subtracts two values (value1 − value2)
- **/** – Divides two values (value1 / value2)
- * – Multiplies two values (value1 * value2)
- **%** – Returns the remainder of dividing two values (value1 % value2)

Arithmetic Operators

Operators	Meaning	Example	Result
+	Addition	4+2	6
-	Subtraction	4-2	2
*	Multiplication	4*2	8
/	Division	4/2	2
%	Modulus operator to get remainder in integer division	5%2	1

- **+=** – Adds the second value to the first value.
- **−=** – Subtracts the second value from the first value.
- **%=** – Divides the first value by the second value and assigns the remainder to the first value.
- ***=** – Multiplies the first value by the second value.
- **/=** – Divides the first value by the second value.

```
x += y ;

x -= y ;

a %= b ;

a *= b ;

a /= b ;
```



```
x = x + y

x = x - y

a = a % b

a = a * b

a = a / b
```

ASSIGNMENT OPERATORS

Create an Arduino program that stores and displays the following student information:

- Name: Rashad
- Age: 21
- Birth Year: 2005
- City: Baku
- Driving License: Yes
- License Category: B
- Car: No
- Exam: Microcontroller
- Passing Score: 67.55%
- Student Score: 74.25346446%
- Result: Passed
- The program should display all information on the Serial Monitor using Serial.print() and Serial.println().



LABORATORY ASSIGNMENT 1